

0.1 Setting Up Coordinates

Place $B = (0, 0)$, $C = (17, 0)$. If $A = (x, y)$, then $x^2 + y^2 = 676$ and $(x - 17)^2 + y^2 = 625$. Subtracting gives $34x - 289 = 51$, so $x = 10$, $y = 24$. Thus

$$A = (10, 24), \quad B = (0, 0), \quad C = (17, 0).$$

Feet of altitudes. Since BC is horizontal, $D = (10, 0)$. Projecting B onto AC and C onto AB :

$$E = \left(\frac{9792}{625}, \frac{2856}{625}\right), \quad F = \left(\frac{425}{169}, \frac{1020}{169}\right).$$

Orthocentre. The altitude from A is $x = 10$; the altitude from B is $y = \frac{7}{24}x$. Hence

$$H = \left(10, \frac{35}{12}\right).$$

Circumcircle. $\Omega : x^2 + y^2 - 17x - \frac{253}{12}y = 0$, with centre $O = \left(\frac{17}{2}, \frac{253}{24}\right)$ and $R = \frac{325}{24}$.

The triangle is acute (e.g. $\cos A, \cos B, \cos C > 0$ by the side lengths), so H lies **inside** Ω .

0.2 Locating X and Q

By Ceva on the concurrent altitudes and Menelaus on transversal E, F, X ,

$$\frac{BX}{XC} = \frac{BD}{DC} = \frac{10}{7} \quad (X \text{ outside } BC),$$

so $X = \left(\frac{170}{3}, 0\right)$ (one checks directly that X lies on line EF).

Line AX has direction $(35, -18)$. Substituting $(10 + 35t, 24 - 18t)$ into Ω :

$$1549t^2 - \frac{759}{2}t = 0 \Rightarrow t = \frac{759}{3098}.$$

Therefore

$$Q = \left(\frac{57545}{3098}, \frac{60690}{3098}\right).$$

0.3 Two Homothety Reductions

The nine-point circle (circle DEF) is the image of Ω under the homothety h centred at H with ratio $\frac{1}{2}$.

Identifying P . Since $h(Q) = \frac{H+Q}{2}$ lies on circle DEF , the midpoint of HQ is on it. The line HQ meets Ω at Q and a second point Q' on the opposite side of H (as H is interior), so the two intersections with circle DEF are the midpoints of HQ and HQ' ; only the former lies on **segment** HQ . Hence

$$P = \frac{H+Q}{2}, \quad PQ = \frac{1}{2}HQ.$$

Bringing in E . Let $E = 2E - H$, the reflection of H in E . Because $HE \perp AC$, E is the reflection of H over line AC , which lies on Ω . Since $E = \frac{H+E}{2}$,

$$P - E = \frac{H+Q}{2} - \frac{H+E}{2} = \frac{Q-E}{2} \Rightarrow PE = \frac{1}{2}QE.$$

Therefore

$$\boxed{\frac{PQ}{PE} = \frac{HQ}{QE}}.$$

0.4 Computing the Two Lengths

Length HQ . With $Q - H = \left(\frac{26565}{3098}, \frac{309925}{18588}\right)$ and $\frac{26565}{3098} = \frac{159390}{18588}$:

$$HQ^2 = \frac{159390^2 + 309925^2}{18588^2}.$$

Now $159390 = 18 \cdot 8855$, $309925 = 35 \cdot 8855$, $18588 = 12 \cdot 1549$, and $18^2 + 35^2 = 1549$:

$$HQ^2 = \frac{8855^2 \cdot 1549}{144 \cdot 1549^2} = \frac{8855^2}{144 \cdot 1549} \Rightarrow HQ = \frac{8855}{12\sqrt{1549}}.$$

Length QE . Here $E = {}^{2E-H} = \left(\frac{13334}{625}, \frac{46669}{7500}\right)$. Over the common denominator 11617500:

$$Q - E = \left(\frac{-32058642}{11617500}, \frac{155297219}{11617500}\right).$$

Factor $32058642 = 12397 \cdot 2586$ and $155297219 = 12397 \cdot 12527$, where $12397 = 7^2 \cdot 11 \cdot 23$. Then

$$32058642^2 + 155297219^2 = 12397^2(2586^2 + 12527^2) = 12397^2 \cdot 163613125,$$

and $163613125 = 5^4 \cdot 13^2 \cdot 1549$ with $12397 \cdot 13 = 161161$, so the numerator equals $161161^2 \cdot 5^4 \cdot 1549$. Since $11617500^2 = 2^4 \cdot 3^2 \cdot 5^8 \cdot 1549^2$,

$$QE^{*2} = \frac{161161^2 \cdot 5^4 \cdot 1549}{2^4 \cdot 3^2 \cdot 5^8 \cdot 1549^2} = \frac{161161^2}{300^2 \cdot 1549} \Rightarrow QE = \frac{161161}{300\sqrt{1549}}.$$

0.5 The Ratio

$$\frac{PQ}{PE} = \frac{HQ}{QE} = \frac{8855/(12\sqrt{1549})}{161161/(300\sqrt{1549})} = \frac{8855 \cdot 25}{161161} = \frac{221375}{161161} = \frac{125}{91},$$

dividing by $1771 = 7 \cdot 11 \cdot 23$ (so $221375 = 125 \cdot 1771$, $161161 = 91 \cdot 1771$).

Since $125 = 5^3$ and $91 = 7 \cdot 13$ are coprime, $a = 125$, $b = 91$, giving

$$a + b = 216.$$